INCH-POUND

MIL-C-24643/20D
22 November 1994
SUPERSEDING
MIL-C-24643/20C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LS7SGU (INCLUDING VARIATION LS7SGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, (see table I for size)
Second - Extruded silicone rubber insulation (see table I for diameter).
Third - Glass braid, providing 80% minimum coverage.
Fourth - Braid covering. Standard identification code applied by method 1
Fifth - The seven conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly. Cabling sequence shall be consecutive starting with no. 1 from center outward.
Sixth - A binder or combination binder and barrier.

Construction variant for type LS7SGU

Seventh - Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant for type LS7SGA

Seventh - Cross-linked polyolefin jacket. (see table I for thickness.
Eighth - Braided metal armor

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TABLE I. Details.

1/ Military part no. M24643/20	Size 2/	Conductor size AWG	Type's overall diameter (inches)				Dia over insulation min (in.)	Dia over braid covering max (in.)	Jacket thickness min avg (in.)	Conductor resistance (ohms)
			LS7SGU		LS7SGA					
			mi	max	min	max				
-01UN	3	16 (Class B)		0.545		.595	0.096	0.130	0.040	4.3
-02UN	4	14 (Class B)		.595		.645	.112	.143	.040	2.68

1/ Unarmored. When armored option is required, see MIL-C-24643 for configuration of part number.

2/ Type LS7SGU and LS7SGA

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EXAMINATION AND TESTS:	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	3000
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Abrasion resistance - minimum	250
Crack resistance -	No damage
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas Flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (maximum for mandrel diameter 12 times maximum cable diameter	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lb/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (insulation) - cycles, minimum	25
* Permanence of printing (jacket) - cycles minimum	125
Cable sealant removability	No failure
Armor (type LS7SGA only) - conformance to material construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Cold working (minus 20 ± 2°C)	No damage
Gas Flame (3 hours)	No failure
Accelerated service - sizes 50 through 800 (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All sizes 2000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
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Agent:
DLA-IS

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